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(54) ELECTRICAL CONNECTION APPARATUS

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(57) **ABSTRACT**

An electrical connection apparatus includes: a probe head holding a probe; a wiring sheet including a first connection portion arranged on a first sheet surface facing the probe head and a second connection portion arranged on a second sheet surface; and a wiring board that is arranged facing the second sheet surface. A press-fit pin having an elastically deforming head is embedded in the wiring sheet, and the head is exposed from at least any of the first sheet surface and the second sheet surface. The head of the press-fit pin exposed from the first sheet surface is fitted into a recess formed in the probe head to fix the probe head to the wiring sheet. The head of the press-fit pin exposed from the second sheet surface is fitted into a recess formed in the wiring board to fix the wiring board to the wiring sheet.

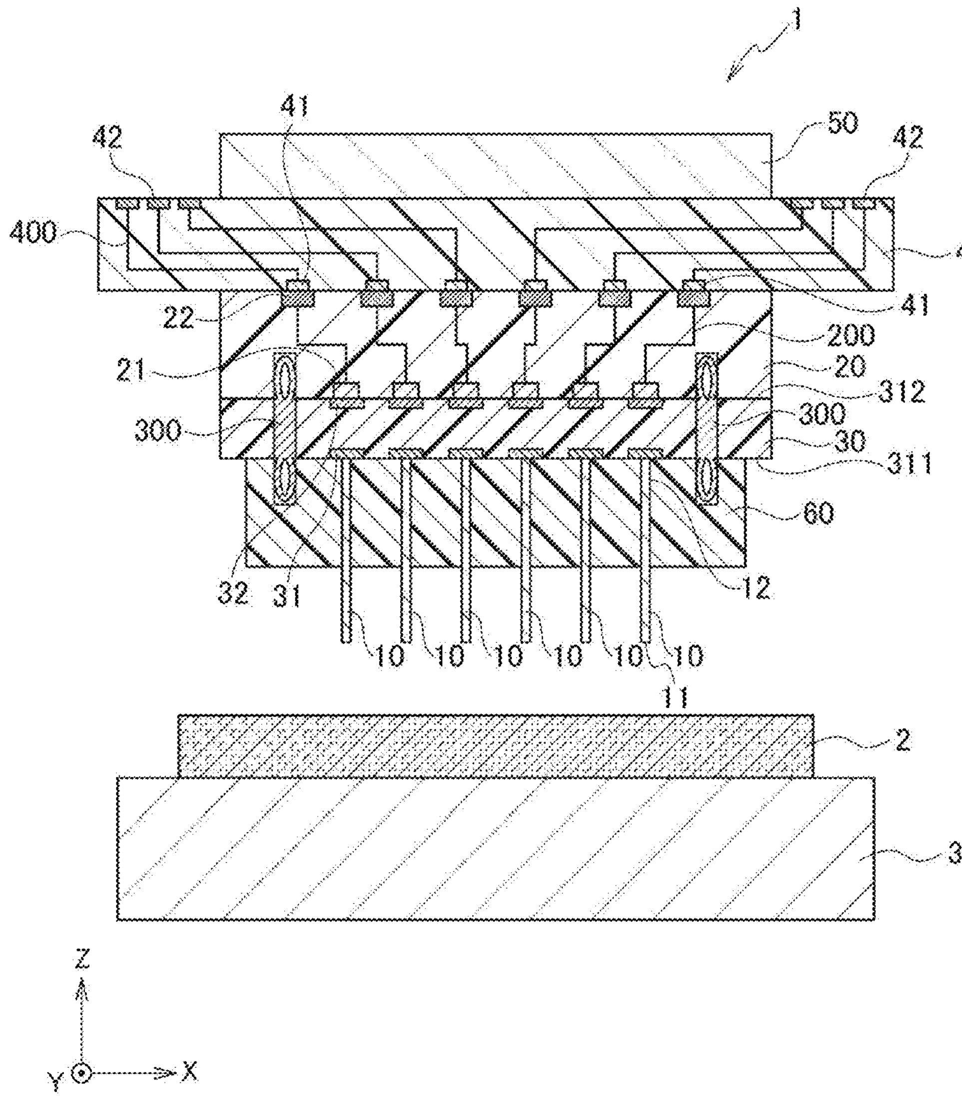


FIG. 1

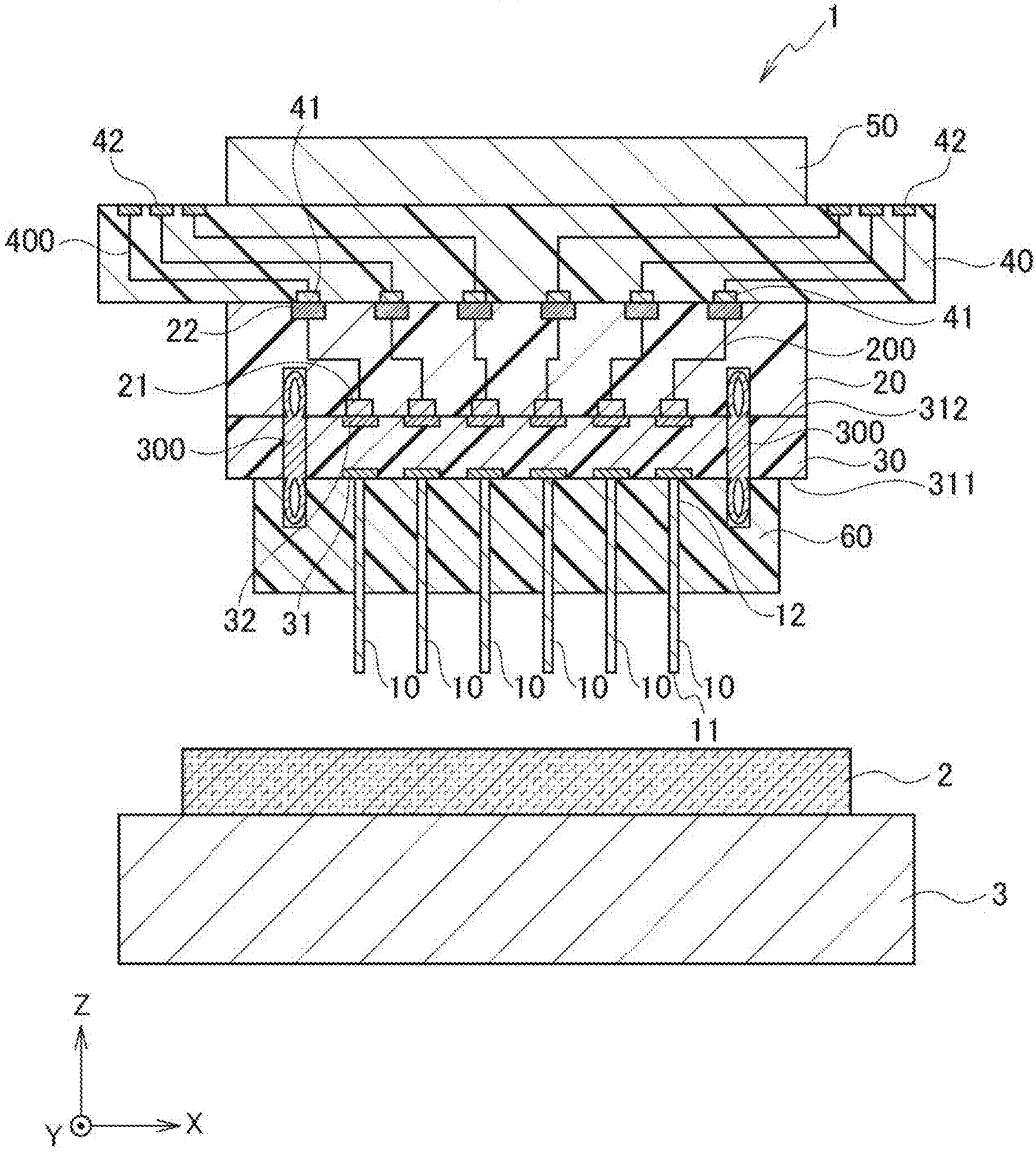

1
41
42
50
42
400
40
41
22
200
21
20
312
300
300
30
311
60
32
31
12
10
10
10
10
10
10
11
2
3
Z
Y
X

FIG. 2

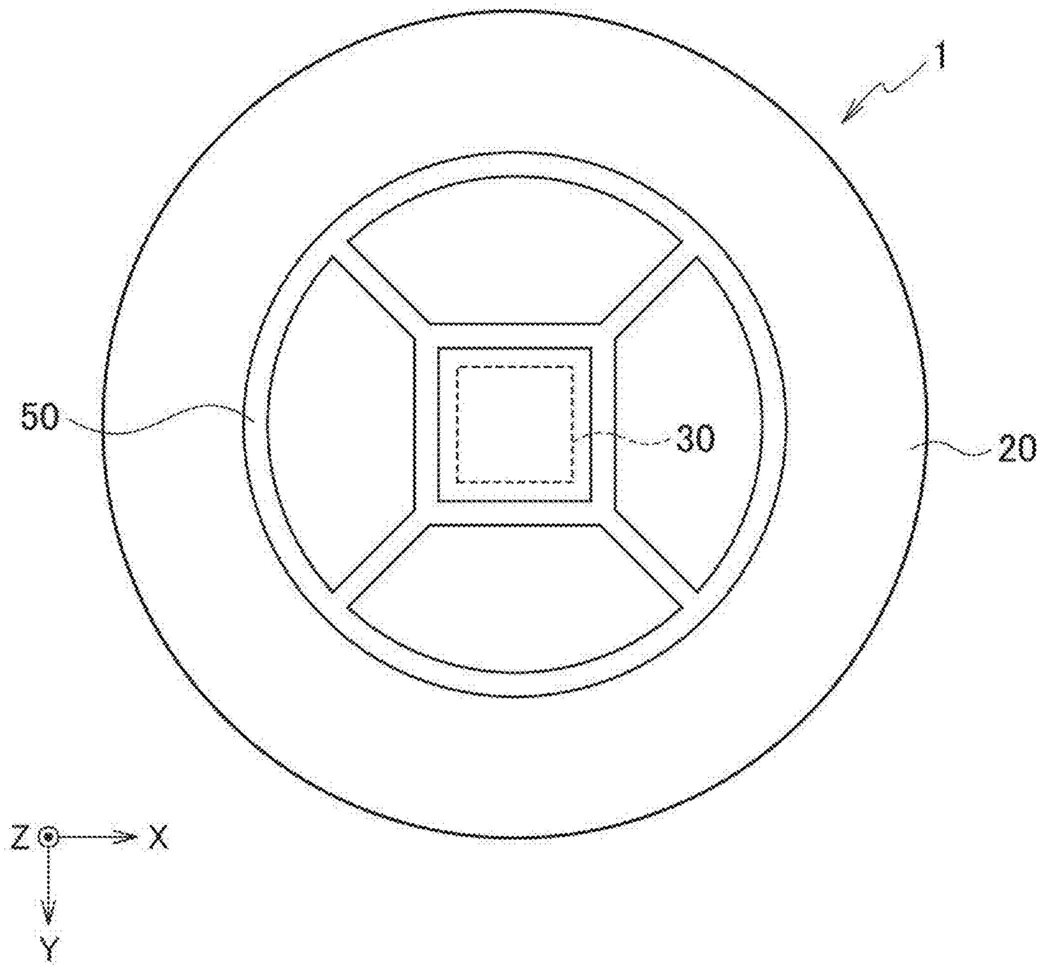


FIG. 3

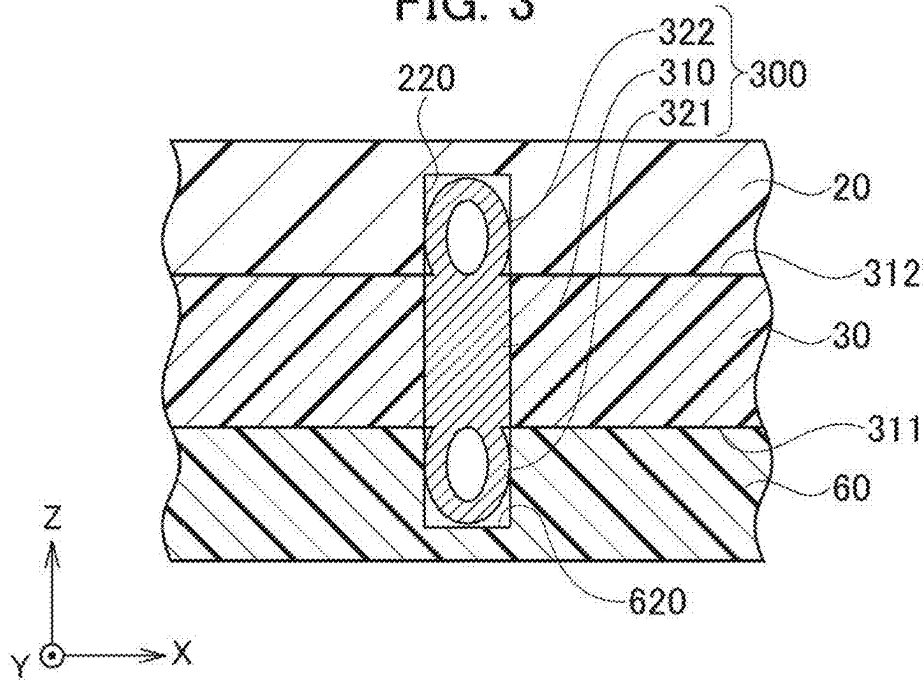

322
310
321
300
220
20
312
30
311
60
620
Z
Y
X

FIG. 4

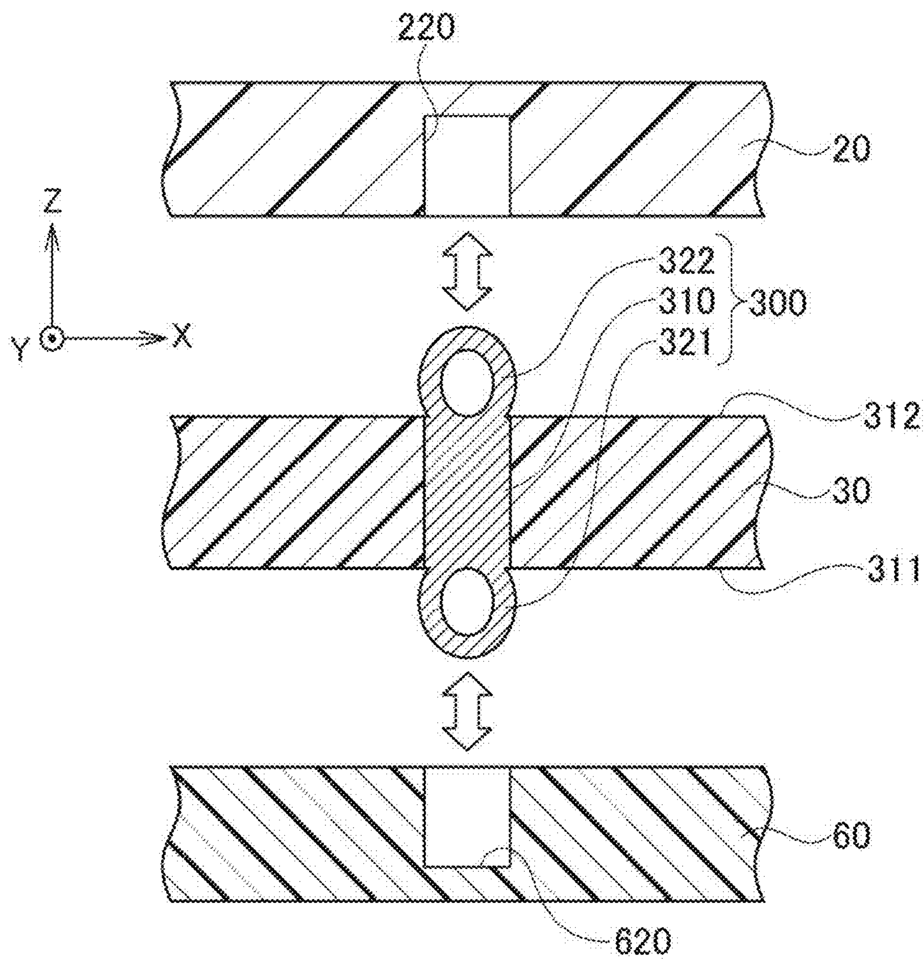

220
20
Z
Y
X
322
310
321
300
312
30
311
60
620

FIG. 5

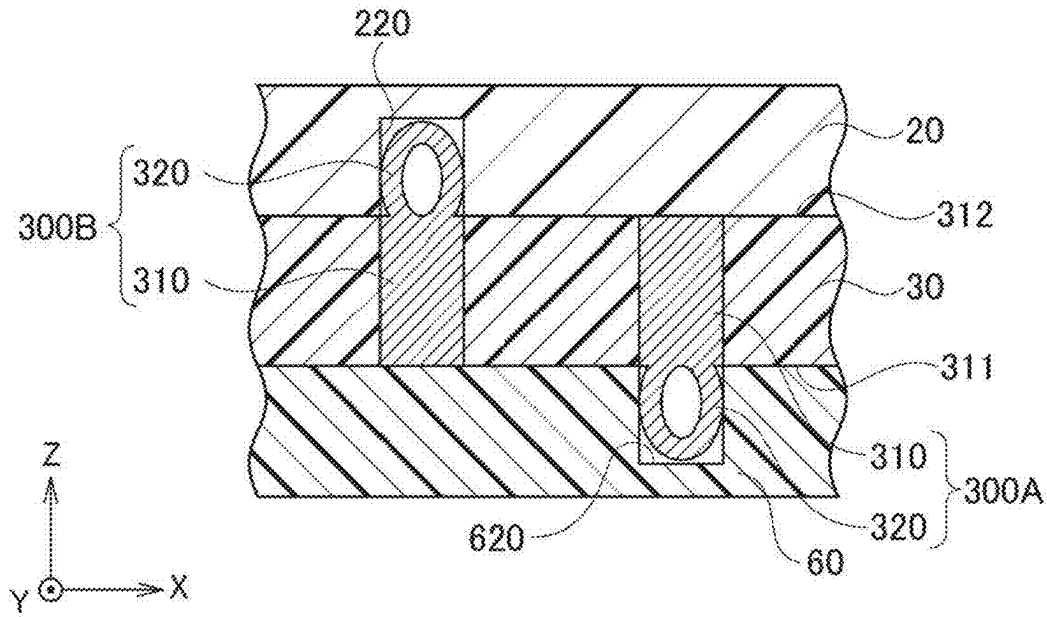

220
20
320
300B
312
310
30
311
310
300A
320
620
60
Z
Y
X

FIG. 6

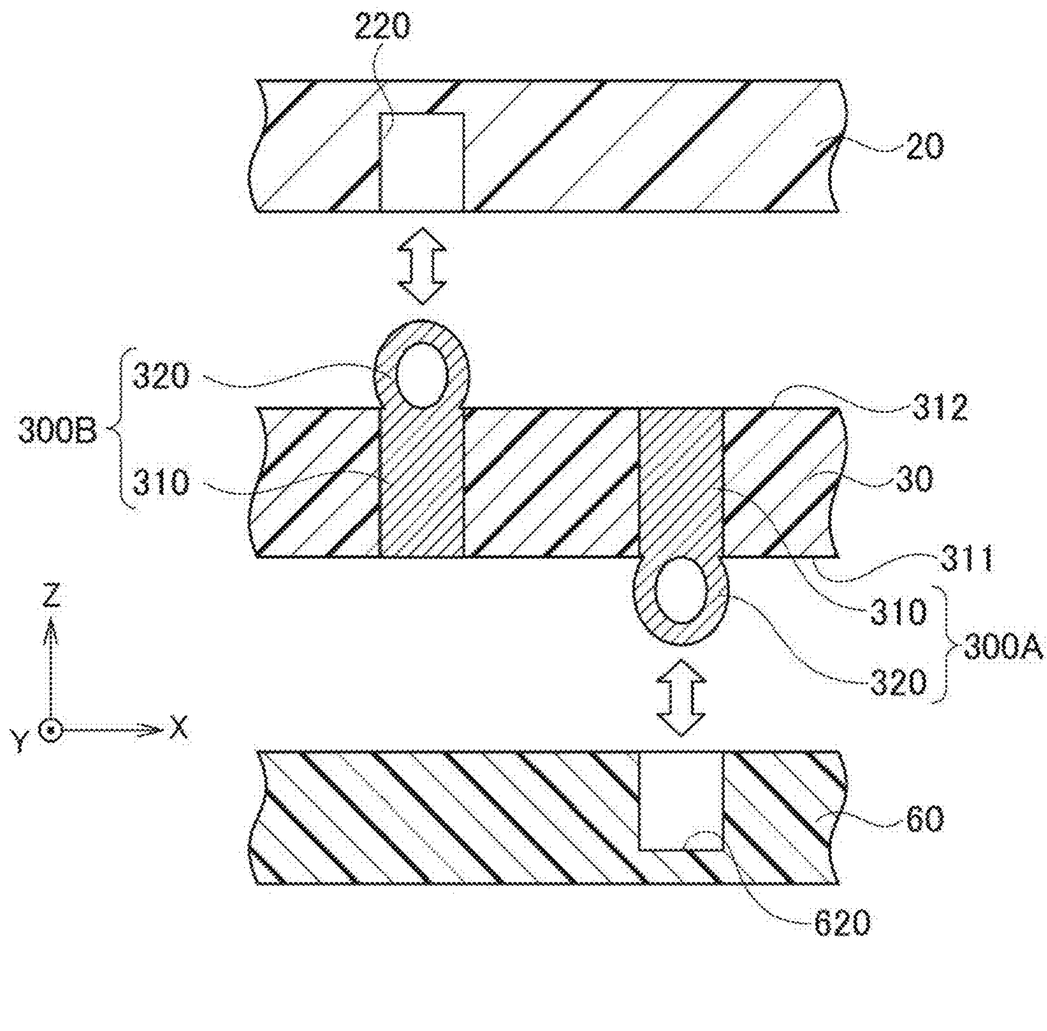

220
20
320
312
300B
310
30
311
310
300A
320
Z
Y
X
60
620

FIG. 7

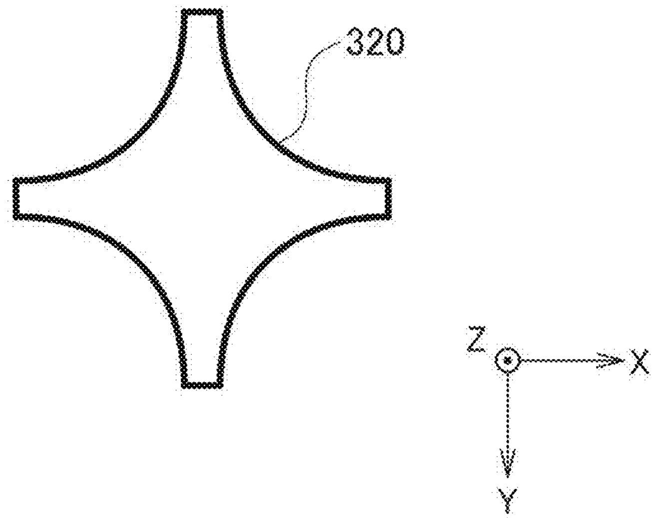


FIG. 8

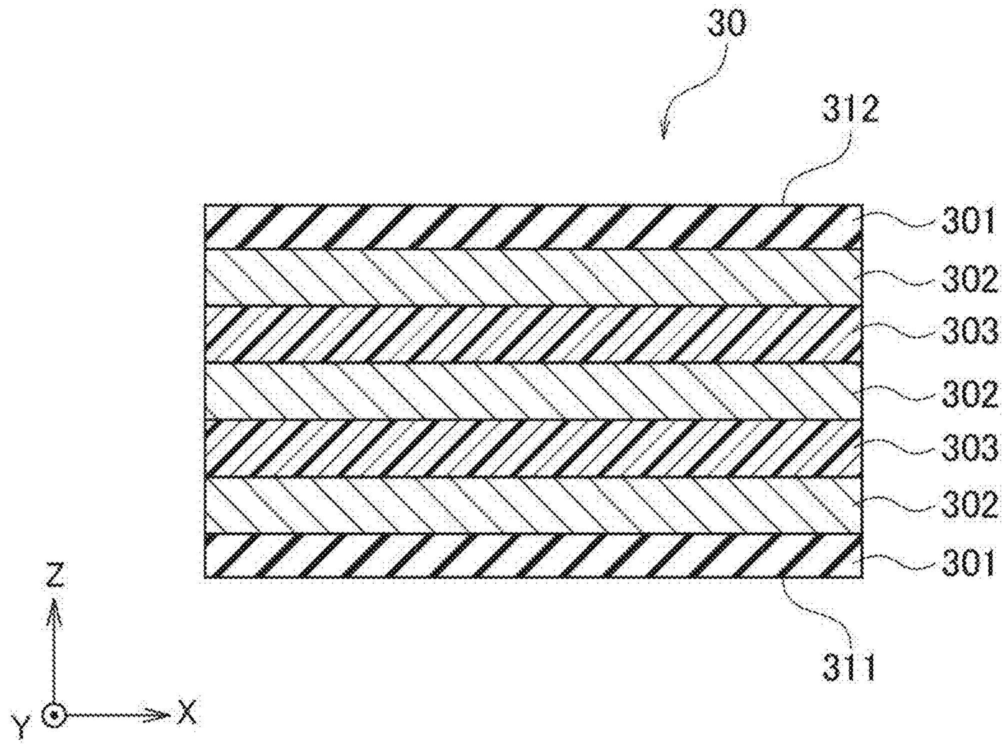


FIG. 9

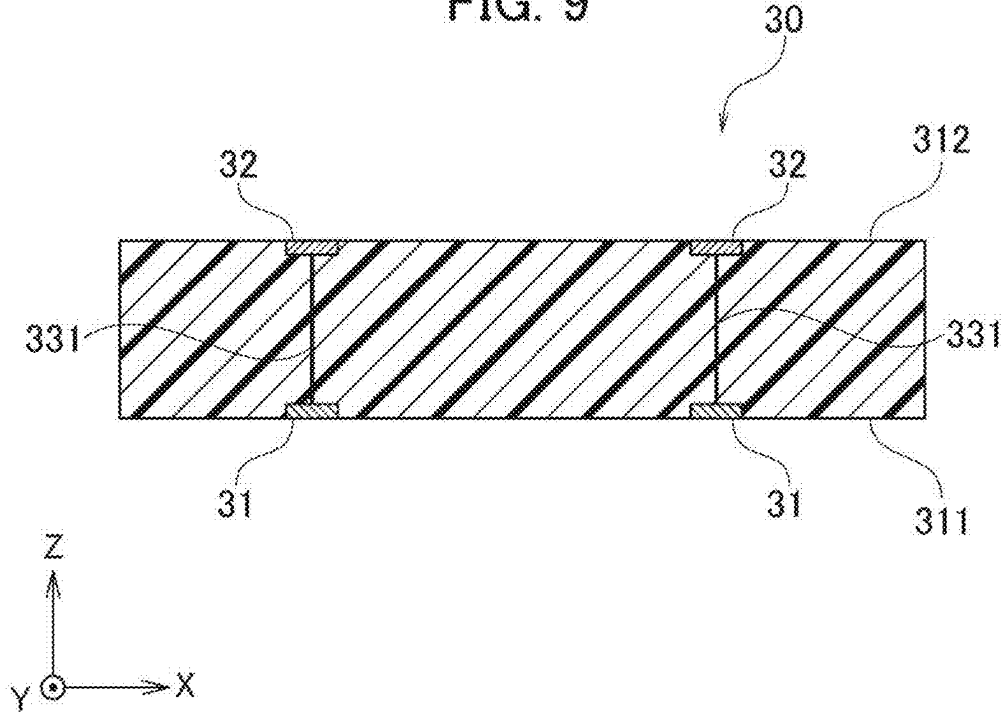

30
32
32
312
331
331
31
31
311
Z
Y
X

FIG. 10

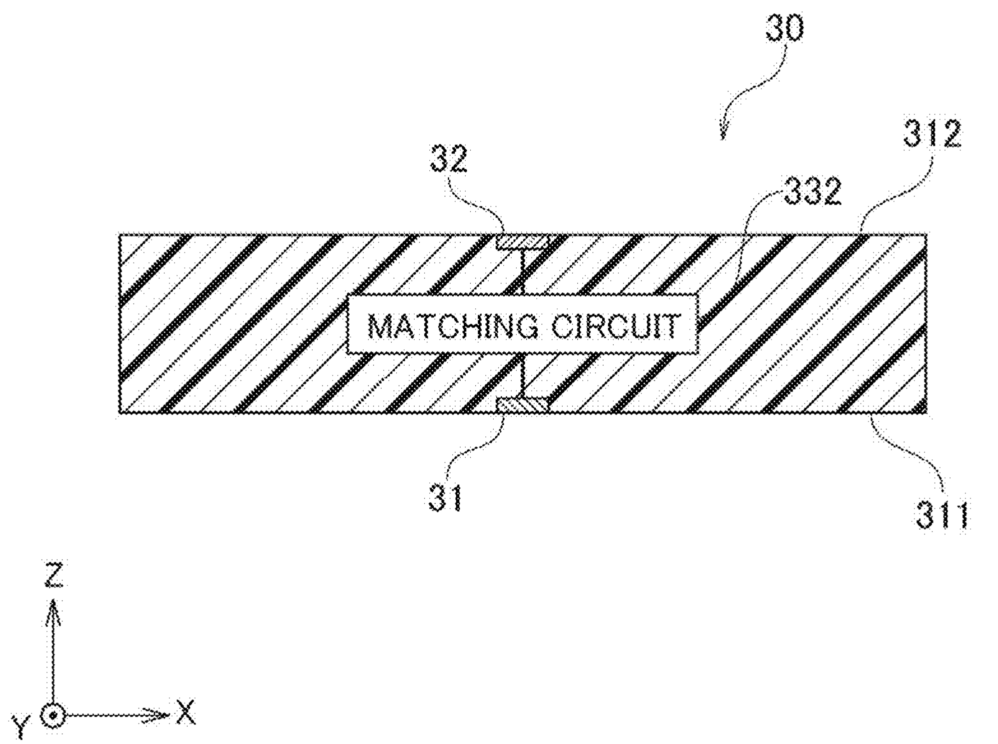

30
32
332
312
MATCHING CIRCUIT
31
311
Z
Y
X

FIG. 11

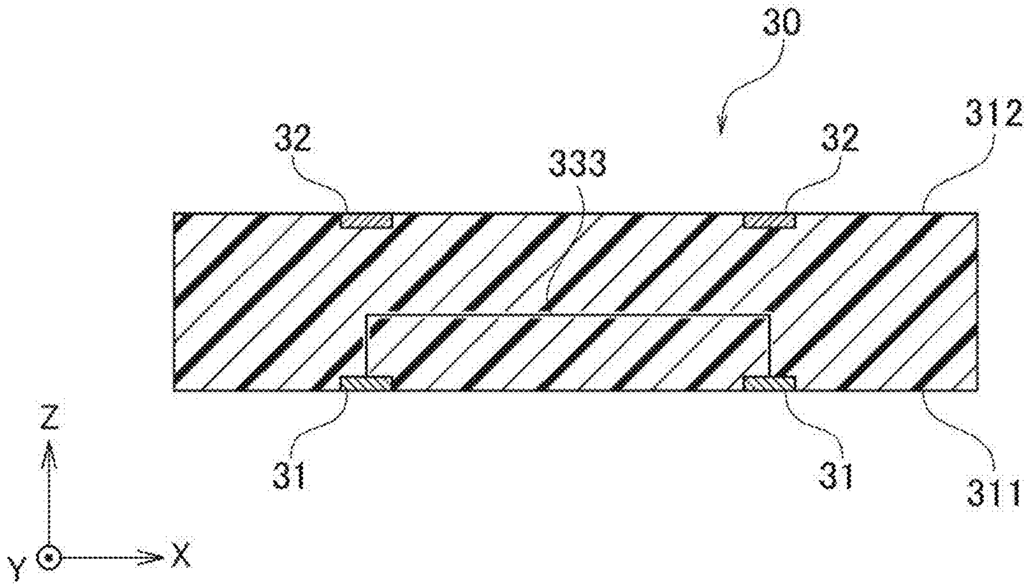

30
32
333
32
312
31
31
311
Z
Y
X

FIG. 12

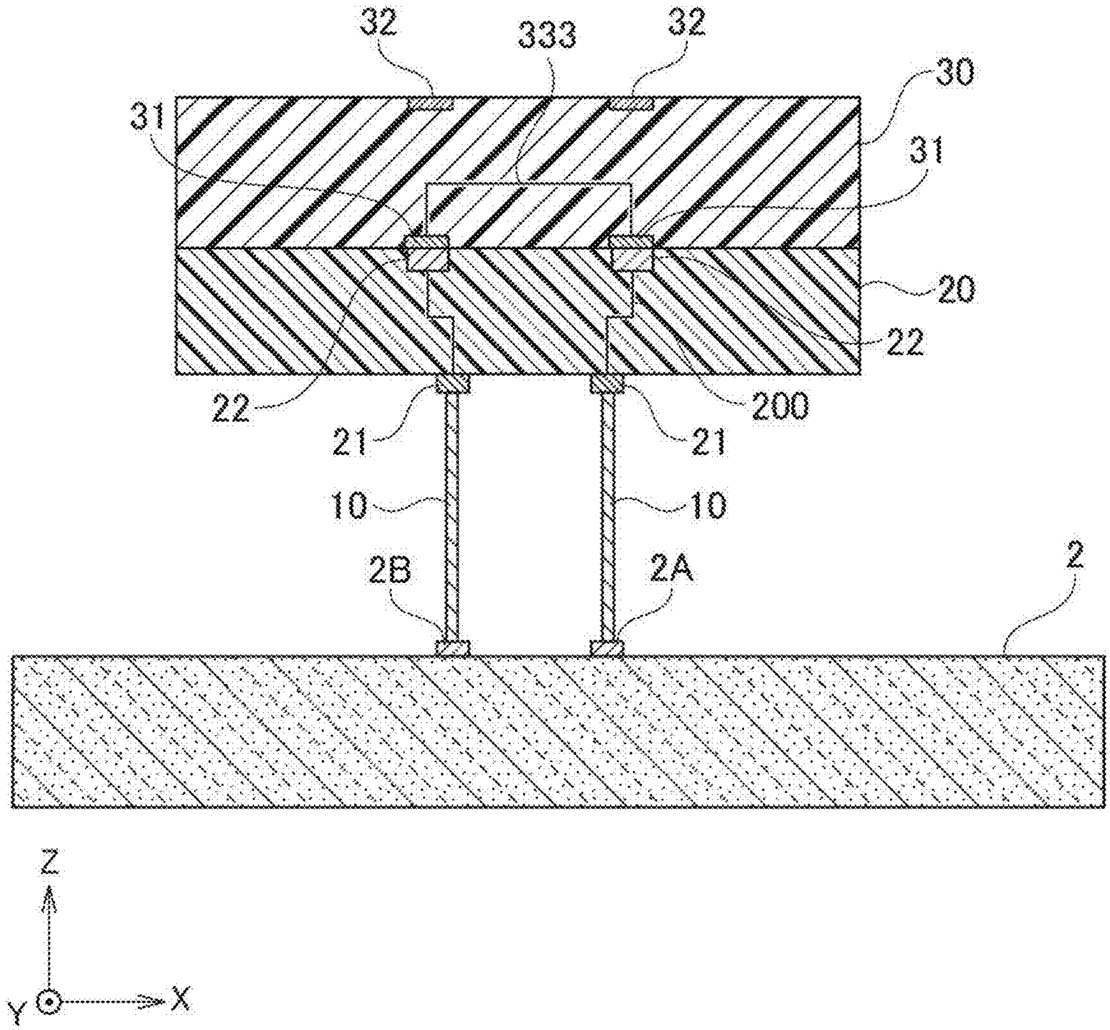

32
333
32
30
31
31
20
22
22
200
21
21
10
10
2B
2A
2
Z
Y
X

FIG. 13

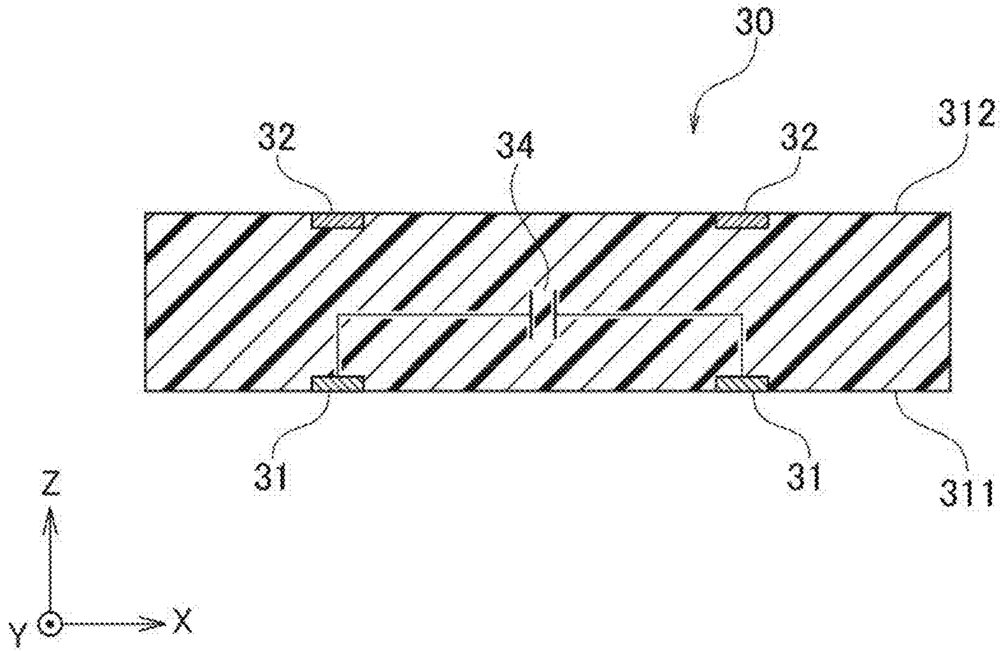

30
32
34
32
312
31
31
311
Z
Y
X

FIG. 14

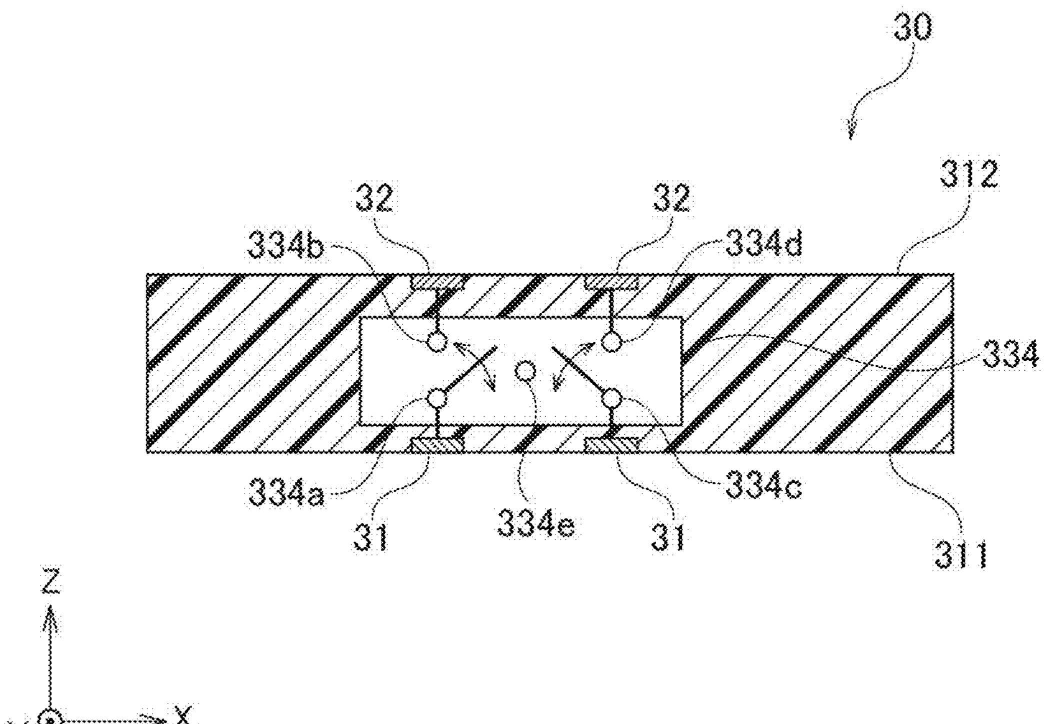

30
32
32
334b
334d
312
334
334a
31
334e
31
334c
311
Z
Y
X

FIG. 15

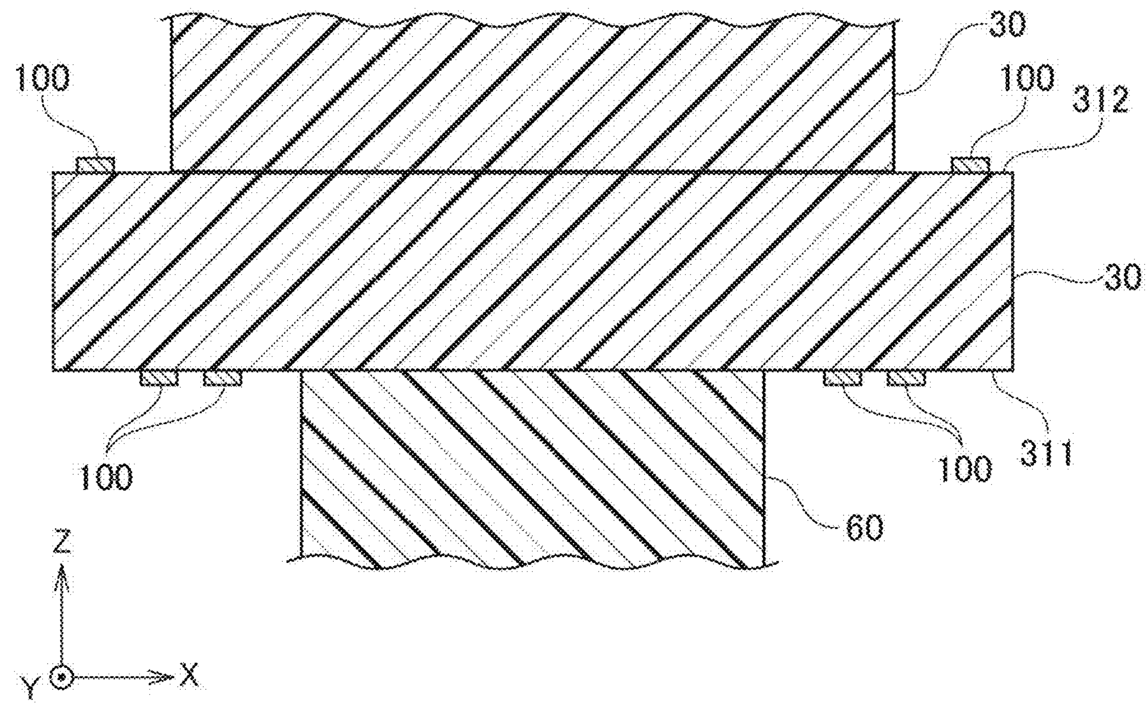
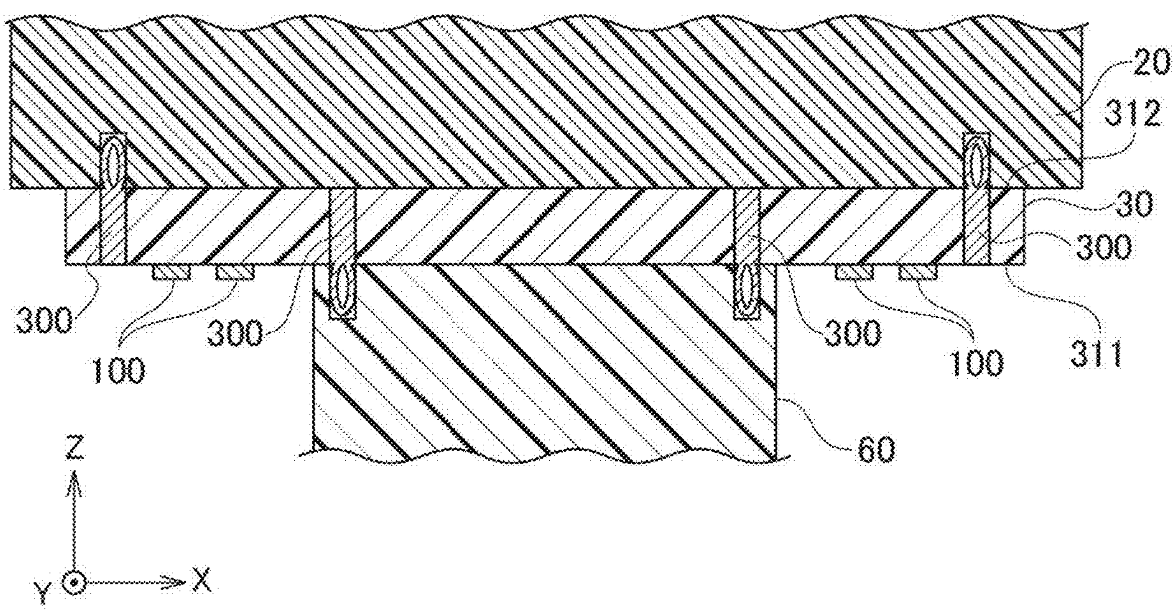

30
100
100
312
30
100
100
311
60
Z
Y
X

FIG. 16


20
312
30
300
300
100
300
300
100
311
60
Z
Y
X

ELECTRICAL CONNECTION APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application is based on, and claims priority from Japanese Patent Application No. 2024-031202, filed on Mar. 1, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an electrical connection apparatus used for inspecting electrical characteristics of an object to be inspected.

BACKGROUND ART

[0003] To measure electrical characteristics of an object to be inspected such as an integrated circuit, an electrical connection apparatus having a terminal for inspection in contact with the object to be inspected is used. In the measurement using the electrical connection apparatus, the object to be inspected is electrically connected to an inspection device such as a tester via the terminal for inspection. In order to measure characteristics of the object to be inspected, the electrical connection apparatus is configured by superimposing a plurality of circuit boards such as a printed board and a space transformer.

CITATION LIST

Patent Literature

[0004] [Patent Literature 1] JP 2019-178901 A

SUMMARY

[0005] Conventionally, a plurality of circuit boards constituting an electrical connection apparatus are screwed and connected. When the circuit boards are connected by means of screwing, it takes time to replace the circuit boards. An object of the present disclosure is to provide an electrical connection apparatus having easily replaceable circuit boards.

[0006] An electrical connection apparatus according to one aspect of the present disclosure includes: a probe head holding a probe; a wiring sheet including a first connection portion arranged on a first sheet surface facing the probe head and a second connection portion arranged on a second sheet surface; and a wiring board that is arranged facing the second sheet surface. A press-fit pin having an elastically deforming head is embedded in the wiring sheet, and the head is exposed from at least any of the first sheet surface and the second sheet surface. The head of the press-fit pin exposed from the first sheet surface is fitted into a recess formed in the probe head to fix the probe head to the wiring sheet. The head of the press-fit pin exposed from the second sheet surface is fitted into a recess formed in the wiring board to fix the wiring board to the wiring sheet.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. **1** is a schematic view showing a configuration of an electrical connection apparatus according to an embodiment.

[0008] FIG. **2** is a schematic plan view showing the configuration of the electrical connection apparatus according to the embodiment.

[0009] FIG. **3** is a schematic view showing a joined state using a press-fit pin of the electrical connection apparatus according to the embodiment.

[0010] FIG. **4** is a schematic view showing a joining method using a press-fit pin of the electrical connection apparatus according to the embodiment.

[0011] FIG. **5** is a schematic view showing another joined state using press-fit pins of the electrical connection apparatus according to the embodiment.

[0012] FIG. **6** is a schematic view showing another joining method using press-fit pins of the electrical connection apparatus according to the embodiment.

[0013] FIG. **7** is a schematic view showing an example of a head shape of a press-fit pin of the electrical connection apparatus according to the embodiment.

[0014] FIG. **8** is a schematic cross-sectional view showing a configuration example of a wiring sheet in the electrical connection apparatus according to the embodiment.

[0015] FIG. **9** is a schematic view showing an example of an internal circuit of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0016] FIG. **10** is a schematic view showing another example of an internal circuit of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0017] FIG. **11** is a schematic view showing another example of an internal circuit of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0018] FIG. **12** is a schematic view showing an example of a loopback circuit using the wiring sheet shown in FIG. **11**.

[0019] FIG. **13** is a schematic view showing another example of an internal circuit of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0020] FIG. **14** is a schematic view showing another example of an internal circuit of the wiring sheet in the electrical connection apparatus according to the embodiment.

[0021] FIG. **15** is a schematic view showing an example of an arrangement of electronic circuits on the wiring sheet in the electrical connection apparatus according to the embodiment.

[0022] FIG. **16** is a schematic view showing another example of a wiring sheet and a circuit board in the electrical connection apparatus according to the embodiment.

DETAILED DESCRIPTION

[0023] An embodiment will be described with reference to the drawings. In the description of the drawings below, the same or similar parts are denoted with the same or similar reference numerals. However, it should be noted that the drawings are schematically shown and the ratios of the thickness of each portion and the like are different from those in reality. Further, it is needless to say that the drawings include portions where the relationships and ratios of dimensions are different between drawings. The following embodiment exemplifies an apparatus and a method for embodying the technical concept of the present invention, and the embodiment does not specify the material, shape, structure, arrangement, and the like of components to the following.

[0024] An electrical connection apparatus **1** according to an embodiment shown in FIG. **1** is used for inspecting an object **2** to be inspected. The electrical connection apparatus **1** includes probes **10**, a probe head **60** holding the probes **10**, a wiring sheet **30** laminated on the probe head **60**, and a wiring board **20** laminated on the wiring sheet **30**.

[0025] The electrical connection apparatus **1** further includes a printed board **40** laminated on the wiring board **20** and a stiffener **50** laminated on the printed board **40**. In the following description, a direction in which the electrical connection apparatus **1** is located when viewed from the object **2** to be inspected is defined as an upward direction, and a direction in which the object **2** to be inspected is located when viewed from the electrical connection apparatus **1** is defined as a downward direction. As shown in FIG. **1**, a direction from the downward direction to the upward direction is defined as a Z direction, and a plane perpendicular to the Z direction is defined as an XY plane. In FIG. **1**, a left-right direction of the drawing is defined as an X direction, and a front-back direction is defined as a Y direction. Further, a surface facing the upward direction of each component of the electrical connection apparatus **1** will be referred to as an upper surface, and a surface facing the downward direction will be referred to as a lower surface.

[0026] The probes **10** function as terminals for inspection for electrically connecting the object **2** to be inspected and an inspection device. Each probe **10** has a tip **11** which is one end arranged so as to be in contact with the object **2** to be inspected and a proximal end **12** which is the other end connected to the tip **11**. The probes **10** are held while passing through through-holes formed in the probe head **60**, for example. Each tip **11** and each proximal end **12** are exposed from the probe head **60**.

[0027] The wiring sheet **30** includes first connection portions **31** arranged on a first sheet surface **311** facing the probe head **60**, and second connection portions **32** arranged on a second sheet surface **312** facing in an opposite direction of the first sheet surface **311**. The first sheet surface **311** is a lower surface of the wiring sheet **30**, and the second sheet surface **312** is an upper surface of the wiring sheet **30**. In other words, the first sheet surface **311** and the second sheet surface **312** are main surfaces of the wiring sheet **30** facing in opposite directions. The wiring sheet **30** has an internal circuit (not shown in FIG. **1**) electrically connected to at least any of the first connection portions **31** and the second connection portions **32**. Each first connection portion **31** is connected to a proximal end **12** of each probe **10** which is exposed from the probe head **60**. The wiring sheet **30** has flexibility and elastically deforms in a thickness direction. Details of the configuration of the wiring sheet **30** will be described later.

[0028] The wiring board **20** is arranged facing the probe head **60** with the wiring sheet **30** therebetween. First electrodes **21** electrically connected to the second connection portions **32** of the wiring sheet **30** are arranged on a lower surface of the wiring board **20** facing the second sheet surface **312** of the wiring sheet **30**. Second electrodes **22** are arranged on an upper surface of the wiring board **20** facing the printed board **40**. The first electrodes **21** and the second electrodes **22** are electrically connected via internal wiring **200**. A multilayer wiring board such as a Multi-Layer Organic (MLO) board or a Multi-Layer Ceramic (MLC) board may be used for the wiring board **20**, for example.

[0029] The printed board **40** is arranged facing the upper surface of the wiring board **20**. The printed board **40** has first ends **41** arranged on a lower surface thereof facing the wiring board **20**, second ends **42** arranged on an upper surface thereof, and wiring patterns **400** for electrically connecting the first ends **41** and the second ends **42**. The second ends **42** are electrically connected to an inspection device (not shown), for example. As a result, an electrical signal propagates between the object **2** to be inspected and the inspection device via the electrical connection apparatus **1**.

[0030] The object **2** to be inspected is mounted on a stage **3**. The electrical connection apparatus **1** and the stage **3** are movable relative to one another in an up-down direction. When the object **2** to be inspected is inspected, a distance between the electrical connection apparatus **1** and the stage **3** is made small, and the tips **11** of the probes **10** contact signal terminals (not shown) of the object **2** to be inspected. FIG. **1** shows a state in which the probes **10** and the object **2** to be inspected are separated.

[0031] The wiring board **20** may be a space transformer that transforms a distance between the proximal ends **12** of the probes **10** to a distance between the first ends **41** of the printed board **40**, when viewed from a direction normal to the upper surface of the wiring board **20**, for example. Due to the wiring board **20** being the space transformer, it is possible to electrically connect the signal terminals arranged on the object **2** to be inspected to the second ends **42** of the printed board **40** which are arranged at a distance larger than a distance between the signal terminals. This facilitates the electrical connection between the wiring patterns **400** in the printed board **40** and the inspection device.

[0032] As shown in FIG. **1**, the stiffener **50** may be laminated on the printed board **40**. The stiffener **50** has higher stiffness than the printed board **40**, and ensures the mechanical strength of the electrical connection apparatus **1** by preventing the printed board **40** from being bent. In addition, the stiffener **50** may be used as a support for fixing each component of the electrical connection apparatus **1**.

[0033] FIG. **2** is a plan view (hereinafter also referred to as "in plan view") of a structure in which the wiring board **20**, the wiring sheet **30**, the printed board **40**, and the stiffener **50** are laminated, as viewed from the Z direction. The stiffener **50** is arranged on an upper surface of the printed board **40** that is circular in plan view.

[0034] As shown in FIG. **2**, the stiffener **50** has a shape in which an outer circular ring and an inner rectangular ring are connected using spokes, for example. The wiring board **20** and the wiring sheet **30** are arranged in the vicinity of the center of a lower surface of the printed board **40**. The wiring board **20** and the printed board **40** are fixed with screws, and the printed board **40** and the stiffener **50** are fixed with screws, for example.

[0035] Meanwhile, as shown in FIG. **1**, press-fit pins **300** are used for joining the probe head **60** and the wiring sheet **30**, and for joining the wiring sheet **30** and the wiring board **20**. Each press-fit pin **300** has heads which elastically deform in a direction perpendicular to an axial direction.

[0036] FIG. **3** shows an example of a joined state using a press-fit pin **300**. The press-fit pin **300** shown in FIG. **3** has a main body **310** passing through the wiring sheet **30**, from the first sheet surface **311** to the second sheet surface **312**. A first head **321** connected to one end of the main body **310** is exposed from the first sheet surface **311**. A second head **322** connected to the other end of the main body **310** is exposed from the second sheet surface **312**. In the following, when each of the first head **321** and the second head **322** is not limited, the first head **321** and the second head **322** will be collectively referred to as a head **320**. The head **320** may be ring-shaped, when viewed from a direction parallel to the XY plane, for example.

[0037] The first head **321** of the press-fit pin **300** exposed from the first sheet surface **311** is fitted into a recess formed in the probe head **60** (hereinafter referred to as "first recess **620**"). This fixes the probe head **60** to the wiring sheet **30**. The second head **322** of the press-fit pin **300** exposed from the second sheet surface **312** is fitted into a recess formed in the wiring board **20** (hereinafter referred to as "second recess **220**"). This fixes the wiring board **20** to the wiring sheet **30**.

[0038] FIG. **4** shows a state in which the probe head **60**, the wiring sheet **30**, and the wiring board **20** are separated. As shown in FIG. **4**, the main body **310** of the press-fit pin **300** is fitted into a through-hole formed in the wiring sheet **30**, for example. At this time, the first head **321** is exposed from the first sheet surface **311**, and the second head **322** is exposed from the second sheet surface **312**. Next, the first sheet surface **311** of the wiring sheet **30** is brought close to the probe head **60** to fit the first head **321** of the press-fit pin **300** into the first recess **620** formed in the probe head **60**. Then, the second sheet surface **312** of the wiring sheet **30** is brought close to the wiring board **20** to fit the second head **322** of the press-fit pin **300** into the second recess **220** formed in the wiring board **20**.

[0039] The order of joining the probe head **60**, the wiring sheet **30**, and the wiring board **20** is arbitrary. The wiring sheet **30** and the wiring board **20** may be joined after joining the probe head **60** and the wiring sheet **30**, for example. Alternatively, the probe head **60** and the wiring sheet **30** may be joined after joining the wiring sheet **30** and the wiring board **20**.

[0040] The first head **321** fitted into the first recess **620** formed in the probe head **60** is compressed and deformed. Due to an elastic force acting on the deformed first head **321** to return a shape thereof to an original shape, the probe head **60** and the wiring sheet **30** are joined. Similarly, the second head **322** fitted into the second recess **220** formed in the wiring board **20** is compressed and deformed. Due to an elastic force acting on the deformed second head **322** to return a shape thereof to an original shape, the wiring board **20** and the wiring sheet **30** are joined.

[0041] FIG. **4** shows a state of a press-fit pin **300** before the electrical connection apparatus **1** is assembled, for example. The press-fit pin **300** shown in FIG. **4** expands in the direction perpendicular to the axial direction, compared with the press-fit pin **300** shown in FIG. **3**.

[0042] FIGS. **3** and **4** show examples of the press-fit pin **300** having the main body **310** embedded in the wiring sheet **30** and the heads **320** exposed from both the first sheet surface **311** and the second sheet surface **312**. Alternatively, one head **320** connected to the main body **310** may be exposed only from either the first sheet surface **311** or the second sheet surface **312**. In other words, the number of head **320** connected to the main body **310** may be one.

[0043] As shown in FIG. **5**, the wiring sheet **30** and the probe head **60** are joined by means of a first press-fit pin **300**A having one head **320** connected to a main body **310**, for example. The head **320** of the first press-fit pin **300**A is exposed from the first sheet surface **311**. The wiring sheet **30** and the wiring board **20** may be joined by means of a second press-fit pin **300**B having one head **320** connected to a main body **310**. The head **320** of the second press-fit pin **300**B is exposed from the second sheet surface **312**.

[0044] As shown in FIG. **5**, overlapping of positions of the first press-fit pin **300**A and the second press-fit pin **300**B is not necessary, in plan view viewed from a direction normal to a main surface of the wiring sheet **30**. In other words, it is not necessary for a position of the press-fit pin **300** for joining the wiring sheet **30** and the probe head **60**, to overlap a position of the press-fit pin **300** for joining the wiring sheet **30** and the wiring board **20** in plan view.

[0045] FIG. **6** shows a state in which the probe head **60**, the wiring sheet **30**, and the wiring board **20** shown in FIG. **5** are separated. As shown in FIG. **6**, the main body **310** of the first press-fit pin **300**A is fitted into a through-hole formed in the wiring sheet **30**. At this time, the head **320** of the first press-fit pin **300**A is exposed from the first sheet surface **311**. Further, the main body **310** of the second press-fit pin **300**B is fitted into a through-hole formed in the wiring sheet **30**. At this time, the head **320** of the second press-fit pin **300**B is exposed from the second sheet surface **312**. Next, the first sheet surface **311** of the wiring sheet **30** is brought close to the probe head **60** to fit the head **320** of the first press-fit pin **300**A into the first recess **620** formed in the probe head **60**. Then, the second sheet surface **312** of the wiring sheet **30** is brought close to the wiring board **20** to fit the head **320** of the second press-fit pin **300**B into the second recess **220** formed in the wiring board **20**. As a result, the probe head **60** and the wiring sheet **30** are fixed, and the wiring board **20** and the wiring sheet **30** are fixed by an elastic force of the head **320**.

[0046] When the press-fit pin **300** has two heads **320**, one of the heads **320** is fitted into one of the first recess **620** and the second recess **220**, and then the other of the heads **320** is fitted into the other of the recesses. Therefore, there is a possibility that the other head **320** which is fitted into the other recess later than the one head may become slightly thin before being fitted. If the thinned head **320** is fitted into the recess, a joining force deteriorates. Due to the press-fit pin **300** having one head **320**, the widely expanded head **320** can be fitted into the second recess **220** and the first recess **620**.

[0047] Further, by using the press-fit pin **300** having one head **320**, the degrees of freedom of an arrangement position of the first recess **620** formed in the probe head **60** and an arrangement position of the second recess **220** formed in the wiring board **20** increase. The second recess **220** can be formed by avoiding positions where the first electrodes **21** and the second electrodes **22** of the wiring board **20** are arranged and positions where the pieces of internal wiring **200** are arranged, for example.

[0048] An arrangement position of the press-fit pin **300** can be arbitrarily set to the extent of not affecting functions of the electrical connection apparatus **1**. Press-fit pins **300** may be arranged at four corners of a rectangular wiring sheet **30** in plan view, for example. Further, press-fit pins **300** may be arranged at any position in addition to the four corners of the wiring sheet **30**.

[0049] Any material may be used for the press-fit pin **300**, but a metal material may be used for the press-fit pin **300**, for example. By using the press-fit pin **300** made of a metal material, the wiring sheet **30** can be joined to the probe head **60** and the wiring board **20** with a strong holding force. However, an insulating material may be used for the press-fit pin **300**. When it is desired to ensure electrical insulation between the wiring sheet **30** and the probe head **60**, or electrical insulation between the wiring sheet **30** and the wiring board **20**, an insulating material may be used for the press-fit pin **300**, for example.

[0050] The head **320** of the press-fit pin may have any shape in plan view. When viewed from an axial direction of the press-fit pin **300**, the head **320** thereof may be circular or star-shaped as shown in FIG. **7**, for example.

[0051] When a plurality of press-fit pins **300** are arranged in the wiring sheet **30**, the maximum value of a diameter perpendicular to an axial direction of each of heads **320** (hereinafter also referred to as "maximum diameter") may be the same for all heads **320**. Maximum diameters of all heads **320** exposed from the first sheet surface **311** may be the same, for example. Further, maximum diameters of all heads **320** exposed from the second sheet surface **312** may be the same. Heads **320** exposed from the first sheet surface **311** of the wiring sheet **30** and heads **320** exposed from the second sheet surface **312** may have the same maximum diameter.

[0052] Meanwhile, a head **320** exposed from the first sheet surface **311** and a head **320** exposed from the second sheet surface **312** may have different maximum diameters. A maximum diameter of the head **320** exposed from the first sheet surface **311** may be made larger than a maximum diameter of the head **320** exposed from the second sheet surface **312**, for example. Therefore, when inner diameters of the first recess **620** and the second recess **220** are the same, a joining force generated between the first recess **620** and the head **320** becomes larger than a joining force generated between the second recess **220** and the head **320**. As a result, when the wiring board **20** is removed from the wiring sheet **30**, heads **320** of all press-fit pins **300** can be fitted into the probe head **60**. In other words, this eliminates a press-fit pin **300** having a head **320** fitted into the wiring board **20**.

[0053] Alternatively, a maximum diameter of the head **320** exposed from the second sheet surface **312** may be made larger than a maximum diameter of the head **320** exposed from the first sheet surface **311**. Therefore, when inner diameters of the first recess **620** and the second recess **220** are the same, a joining force generated between the second recess **220** and the head **320** becomes larger than a joining force generated between the first recess **620** and the head **320**. As a result, when the probe head **60** is removed from the wiring sheet **30**, heads **320** of all press-fit pins **300** can be fitted into the wiring board **20**. In other words, this eliminates a press-fit pin **300** having a head **320** fitted into the probe head **60**.

[0054] As described above, due to the head **320** exposed from the first sheet surface **311** and the head **320** exposed from the second sheet surface **312** having different maximum diameters, heads **320** of press-fit pins **300** can be fitted into either the probe head **60** or the wiring board **20**. This can suppress the wiring sheet **30** from being pulled in two directions by a press-fit pin **300** having a head **320** fitted into the probe head **60** and a press-fit pin **300** having a head **320** fitted into the wiring board **20**, and also can suppress films formed on surfaces of the wiring sheet **30** from being peeled off.

[0055] Further, when the head **320** exposed from the first sheet surface **311** and the head **320** exposed from the second sheet surface **312** have the same maximum diameter, an inner diameter of the first recess **620** formed in the probe head **60** and an inner diameter of the second recess **220** formed in the wiring board **20** may be different. The inner diameter of the first recess **620** may be made smaller than the inner diameter of the second recess **220**, for example. Therefore, a joining force generated between the first recess **620** and the head **320** becomes larger than a joining force generated between the second recess **220** and the head **320**. As a result, when the wiring board **20** is removed from the wiring sheet **30**, heads **320** of all press-fit pins **300** can be fitted into the probe head **60**.

[0056] Alternatively, an inner diameter of the second recess **220** may be made smaller than an inner diameter of the first recess **620**. Therefore, a joining force generated between the second recess **220** and the head **320** becomes larger than a joining force generated between the first recess **620** and the head **320**. As a result, when the probe head **60** is removed from the wiring sheet **30**, heads **320** of all press-fit pins **300** can be fitted into the wiring board **20**.

[0057] As described above, due to the first recess **620** and the second recess **220** having different inner diameters, heads **320** of press-fit pins **300** can be fitted into either the probe head **60** or the wiring board **20**. This can suppress the wiring sheet **30** from being pulled in two directions by a press-fit pin **300** having a head **320** fitted into the probe head **60** and a press-fit pin **300** having a head **320** fitted into the wiring board **20**, and also can suppress films formed on surfaces of the wiring sheet **30** from being peeled off.

[0058] The wiring sheet **30** may have a structure in which a conductive film and an insulating film are laminated. An internal circuit may be constituted from a conductive pattern formed on a conductive film, for example. FIG. **8** shows an example of a configuration of the wiring sheet **30**. The wiring sheet **30** shown in FIG. **8** has a structure in which a laminated body constituted by conductive films **302** and insulating films **303** is interposed between a pair of cover films **301** of insulating materials. The number of laminated bodies constituted by the conductive films **302** and the insulating films **303** can be set to any number. The conductive films **302** may be metal materials such as copper foils, for example. The insulating films **303** may be insulating materials such as polyimide sheets, for example. The cover films **301** may be insulating materials such as solder resists, for example. An adhesive may be used for joining the conductive films **302**, the insulating films **303**, and the cover films **301** each other, for example. The wiring sheet **30** may have a structure in which a film (also referred to as a "base film") of an insulating material is further interposed between laminated bodies including the conductive films **302** and the insulating films **303**.

[0059] The wiring sheet **30** may be selected from among a plurality of wiring sheet candidates. Each of the wiring sheet candidates has a first connection portion **31** arranged on the first sheet surface **311**, a second connection portion **32** arranged on the second sheet surface **312**, and an internal circuit. Each of the wiring sheet candidates may include an internal circuit having a configuration different from that of another wiring sheet candidate. One wiring sheet **30** selected from among the plurality of wiring sheet candidates may be configured in an attachable/detachable manner between the probe head **60** and the wiring board **20**.

[0060] An example of the configuration of wiring sheet candidates included in a wiring sheet group will be described below. In the following, when each of the wiring sheet candidates is not limited, the wiring sheet candidates will be collectively referred to as a wiring sheet **30**.

[0061] An internal circuit of a wiring sheet **30** as any one of the group of wiring sheets may include a circuit for electrically connecting the first connection portions **31** and the second connection portions **32** (hereinafter also referred to as an "interposer circuit"). When the interposer circuit is formed on the wiring sheet **30** having the structure shown in FIG. **8**, wiring is formed, which passes through the cover films **301**, the conductive films **302**, and the insulating films **303**, from the first sheet surface **311** to the second sheet surface **312** of the wiring sheet **30**, for example.

[0062] If the internal circuit of the wiring sheet **30** includes the interposer circuit, the internal circuit of the wiring sheet **30** shown in FIG. **9** may include a circuit for short-circuiting the first connection portions **31** and the second connection portions **32**. In the internal circuit of the wiring sheet **30** shown in FIG. **9**, the first connection portions **31** and the second connection portions **32** are electrically short-circuited by short-circuit wiring **331**.

[0063] By attaching the wiring sheet **30** shown in FIG. **9** to the electrical connection apparatus **1**, the probes **10** and the wiring patterns **400** in the printed board **40** are short-circuited via the internal circuit of the wiring sheet **30**. This electrically connects the object **2** to be inspected and the inspection device. As a result, an electrical signal propagates between an inspection device such as an IC tester and the object **2** to be inspected and characteristics of the object **2** to be inspected are measured. One first connection portion **31** and one second connection portion **32** are connected in a one-to-one relationship by means of the internal circuit of the wiring sheet **30**, for example. Alternatively, one first connection portion **31** may be connected to a plurality of second connection portions **32**, or a plurality of first connection portions **31** may be connected to one second connection portion **32** by means of the internal circuit.

[0064] When the internal circuit of the wiring sheet **30** includes the interposer circuit, the internal circuit may include a matching circuit **332** having a first terminal connected to a first connection portion **31** and a second terminal connected to a second connection portion **32** as shown in FIG. **10**. The matching circuit **332** may perform impedance matching between the first connection portion **31** and the second connection portion **32**. The matching circuit **332** may include a x-type filter, for example.

[0065] The internal circuit of the wiring sheet **30** may include a circuit for electrically connecting any one of the first connection portions **31** and another one of the first connection portions **31**. In other words, the internal circuit of the wiring sheet **30** may include a circuit for electrically connecting an output terminal and an input terminal of the object **2** to be inspected (hereinafter also referred to as a "loopback circuit"). Two signal terminals of the object **2** to be inspected are electrically connected by the loopback circuit.

[0066] The internal circuit of the wiring sheet **30** may include a circuit for short-circuiting any one of the first connection portions **31** and another one of the first connection portions **31** as shown in FIG. **11**, for example. In the internal circuit of the wiring sheet **30** shown in FIG. **11**, the one first connection portion **31** and the other first connection portion **31** are electrically short-circuited by loopback wiring **333**. By attaching the wiring sheet **30** shown in FIG. **11** to the electrical connection apparatus **1**, one of the probes **10** and another one of the probes **10** are short-circuited via the internal circuit of the wiring sheet **30**. This electrically connects one signal terminal and the other signal terminal of the object **2** to be inspected.

[0067] FIG. **12** shows a configuration in which a first signal terminal **2**A of the object **2** to be inspected in contact with one of the probes **10**, and a second signal terminal **2**B of the object **2** to be inspected in contact with another one of the probes **10** are electrically connected via the loopback wiring **333** of the wiring sheet **30**. Suppose that the object **2** to be inspected is a receiving circuit, the first signal terminal **2**A is an output terminal of the object **2** to be inspected, and the second signal terminal **2**B is an input terminal of the object **2** to be inspected, for example. In the above case, a transmission test can be performed by returning an output from the object **2** to be inspected to an input. In other words, it is possible to test whether an output portion and an input portion of the object **2** to be inspected are functioning normally, even if there is no destination equipment for transmission. As an inspection in accordance with a jitter tolerance test performed for the receiving circuit, an output signal output from the first signal terminal **2**A (output terminal) may be input to the second signal terminal **2**B (input terminal) as an input signal to inspect whether a specified error rate is ensured, for example.

[0068] When the internal circuit of the wiring sheet **30** includes the loopback circuit, the internal circuit may include a circuit having a capacitor **34** connected in series between any one of the first connection portions **31** and another one of the first connection portions **31** as shown in FIG. **13**. One terminal of the capacitor **34** is connected to the one first connection portion **31**, and the other terminal of the capacitor **34** is connected to the other first connection portion **31**. The capacitor **34** may be a capacitor formed using a semiconductor manufacturing process (hereinafter also referred to as a "process capacitor").

[0069] Further, the internal circuit of the wiring sheet **30** may include a relay circuit that switches to electrically connect one of the first connection portions **31** to either one of the second connection portions **32** or another one of the first connection portions **31**. The internal circuit shown in FIG. **14** includes a relay circuit **334** that constitutes either the interposer circuit for connecting a first connection portion **31** and a second connection portion **32** or the loopback circuit for connecting first connection portions **31** each other, for example.

[0070] When the relay circuit **334** shown in FIG. **14** constitutes the interposer circuit, a first contact terminal **334***a* and a second contact terminal **334***b* are connected, and a third contact terminal **334***c* and a fourth contact terminal **334***d* are connected. This electrically connects a first connection portion **31** and a second connection portion **32**. When the relay circuit **334** constitutes the loopback circuit, the first contact terminal **334***a* and a common contact terminal **334***e* of the relay circuit **334** are connected, and the third contact terminal **334***c* and the common contact terminal **334***e* are connected. This electrically connects one of the first connection portions **31** and another one of the first connection portions **31**.

[0071] Although examples of the internal circuit of the wiring sheet **30** have been described with reference to FIGS.

9 to 14, it is needless to say that the configuration of the internal circuit is not limited to the above. The internal circuit may include an inductor instead of the capacitor **34** shown in FIG. **13**, or the internal circuit may include both a capacitor and an inductor, for example. In other words, the internal circuit of the wiring sheet **30** may include a passive circuit including any element. Further, the internal circuit may include a switching circuit using a diode or the like instead of the relay circuit **334** shown in FIG. **14**.

[0072] An element included in the internal circuit of the wiring sheet **30** may be formed using a Micro Electro Mechanical Systems (MEMS) process, for example. By using the MEMS process, an element which is reduced in size can be formed integrally with the wiring sheet **30**.

[0073] As described above, various circuit configurations as shown in FIGS. **9** to **14** can be implemented in the electrical connection apparatus **1** by merely replacing the wiring sheet **30** in the electrical connection apparatus **1**, for example. Therefore, a plurality of types of measurements can be performed on the object **2** to be inspected using the electrical connection apparatus **1**. A DC test may be performed on the object **2** to be inspected by mounting, in the electrical connection apparatus **1**, the wiring sheet **30** including the internal circuit for short-circuiting the first connection portion **31** and the second connection portion **32**, for example. Further, a high-frequency test may be performed on the object **2** to be inspected by mounting, in the electrical connection apparatus **1**, the wiring sheet **30** including the internal circuit having the matching circuit or the loopback circuit.

[0074] Further, by arranging the relay circuit **334** in the wiring sheet **30** shown in FIG. **14** of the electrical connection apparatus **1**, the wiring length of the interposer circuit and the loopback circuit can be reduced, compared to that when a relay element is arranged on the printed board **40**. As a result, according to the electrical connection apparatus **1**, it is possible to shorten a propagation path of an electrical signal and suppress the loss of an electrical signal and noise.

[0075] Further, by arranging the matching circuit **332** in the wiring sheet **30** shown in FIG. **10** of the electrical connection apparatus **1**, wiring connected to the matching circuit **332** can be shortened. The matching circuit **332** can be arranged in the immediate vicinity of wiring for which impedance matching is to be performed, and therefore impedance matching can be performed effectively, for example.

[0076] As described above, the electrical connection apparatus **1** has the wiring sheet **30** including the internal circuit capable of constituting any circuit, which is interposed between the wiring board **20** and the printed board **40** in an attachable/detachable manner, and therefore it is possible to configure an arbitrary measurement system. As a result, characteristics of the object **2** to be inspected can be measured with high accuracy with the electrical connection apparatus **1**.

[0077] Further, the wiring sheet **30** with flexibility in the thickness direction, which is interposed between the probe head **60** and the wiring board **20** in the electrical connection apparatus **1**, can mitigate warpages and irregularities of surfaces of the probe head **60** and the wiring board **20**. This can suppress the rattling and contact failure of the electrical connection apparatus caused by a gap between the probe head **60** and the wiring board **20**.

[0078] Further, electronic components electrically connected to the internal circuit can be arranged on a surface of the wiring sheet **30** in the electrical connection apparatus **1**. As shown in FIG. **15**, electronic components **100** connected to the internal circuit of the wiring sheet **30** may be arranged on both of the first sheet surface **311** and the second sheet surface **312**, for example. Each electronic component **100** may be a capacitor, an inductor, or a resistance element, for example. FIG. **15** shows an example of arranging the electronic components **100** on both of the first sheet surface **311** and the second sheet surface **312**, but the electronic components **100** may be arranged on either the first sheet surface **311** or the second sheet surface **312**. By arranging the electronic components **100** on a surface of the wiring sheet **30**, it is possible to reduce or divide the number of electronic components arranged on the wiring board **20** or the printed board **40**, for example. Further, by arranging electronic components on the surface of the wiring sheet **30**, it is possible to reduce the wiring length between the object **2** to be inspected and each electronic component. Therefore, by arranging electronic components such as capacitors for reducing power source noise on the wiring sheet **30**, it is possible to stably perform measurements on the object **2** to be inspected, for example.

[0079] When the electronic components **100** are arranged on the surface of the wiring sheet **30**, ends of the wiring sheet **30** that extend outward from the probe head **60** in plan view may hang downward. Therefore, as shown in FIG. **16**, outer edges of the wiring board **20** may be positioned outward from outer edges of the wiring sheet **30** in plan view. By arranging press-fit pins **300** for joining the wiring sheet **30** and the wiring board **20** outside the electronic components **100** arranged on the wiring sheet **30**, it is possible to prevent the ends of the wiring sheet **30** from hanging down. FIG. **16** shows an example of using each press-fit pin **300** having one head **320** for joining the probe head **60** and the wiring sheet **30**, but each press-fit pin **300** having two heads **320** may be used to join the probe head **60** and the wiring sheet **30**.

[0080] As described above, in the electrical connection apparatus **1**, press-fit pins **300** are used for joining the probe head **60** and the wiring sheet **30**, and for joining the wiring board **20** and the wiring sheet **30**. This facilitates attachment and detachment of the probe head **60** to and from the wiring sheet **30** in the electrical connection apparatus **1**, compared to when the probe head **60**, the wiring sheet **30**, and the wiring board **20** are screwed and joined. Therefore, the wiring sheet **30** can be easily replaced.

[0081] In addition, by using a press-fit pin **300** instead of a screw, it is not necessary to perform processing to form a screw hole, and the production of the electrical connection apparatus becomes easy. An inner diameter of each of the first recess **620** and the second recess **220** is formed smaller than an inner diameter of a screw hole. This can reduce the damage caused when the probe head **60** and the wiring board **20** are processed.

[0082] Further, by using the press-fit pin **300**, it is possible to use the second recess **220** having a smaller inner diameter than the screw hole. This enhances the degree of freedom of the arrangement of the electrodes and wiring of the wiring board **20**. The wiring board **20** is likely to trap heat inside by the difference between the inner diameter of the screw hole and the inner diameter of the recess. This can increase the heat capacity.

Other Embodiments

[0083] Although an embodiment of the present invention has been described above, the discussion and drawings forming part of this disclosure should not be construed as limiting the invention. Various alternative embodiments, examples, and operational techniques will be apparent to those skilled in the art from this disclosure.

[0084] The wiring sheet **30** may include an internal circuit in which a plurality of types of circuit configurations are mixed, for example. The wiring sheet **30** may include an internal circuit in which the short-circuit wiring **331** and the matching circuit **332** are mixed, for example. Alternatively, the wiring sheet **30** may include an internal circuit in which an interposer circuit and a loopback circuit are mixed. The internal circuit may include the short-circuit wiring **331**, the matching circuit **332**, and the loopback wiring **333**, or may further include the relay circuit **334**, for example. Further, the loopback circuit may include the matching circuit **332**. In this way, any circuit can be configured in the internal circuit of the wiring sheet **30**.

[0085] As described above, by mounting, in the electrical connection apparatus **1**, the wiring sheet **30** including the internal circuit in which any circuit configurations are mixed, a plurality of types of measurements can be performed on the object **2** to be inspected with one wiring sheet **30**.

[0086] In the case described above, terminals for inspection that contact signal terminals of the object **2** to be inspected are the probes **10**. When the terminals for inspection are the probes **10**, it is possible to measure electrical characteristics of an object **2** to be inspected not separated from a wafer or an object **2** to be inspected which is converted into a chip, for example. Meanwhile, the wiring sheet **30** may be mounted in an electrical connection apparatus including a test socket, to support the measurement of electrical characteristics of the object **2** to be inspected performed in a state where the object **2** to be inspected is mounted in a package or the like. The test socket may be arranged on the wiring sheet **30**, the test socket having a terminal for inspection connected to an external terminal of the package in which the object **2** to be inspected is mounted, instead of the probes **10** in the electrical connection apparatus **1** shown in FIG. **1**, for example.

[0087] In this way, it is needless to say that the present invention includes various embodiments not described above. Therefore, the technical scope of the present invention is defined only by matters specified in the invention that are within the scope of claims appropriate from the above description.

REFERENCE SIGNS LIST

[0088] **1** Electrical connection apparatus
[0089] **2** Object to be inspected
[0090] **10** Probe
[0091] **11** Tip
[0092] **12** Proximal end
[0093] **20** Wiring board
[0094] **21** First electrode
[0095] **22** Second electrode
[0096] **30** Wiring sheet
[0097] **31** First connection portion
[0098] **32** Second connection portion
[0099] **40** Printed board
[0100] **41** First end
[0101] **42** Second end
[0102] **50** Stiffener
[0103] **60** Probe head
[0104] **100** Electronic component
[0105] **200** Internal wiring
[0106] **220** Second recess
[0107] **300** Press-fit pin
[0108] **310** Main body
[0109] **311** First sheet surface
[0110] **312** Second sheet surface
[0111] **320** Head
[0112] **321** First head
[0113] **322** Second head
[0114] **400** Wiring pattern
[0115] **620** First recess

What is claimed is:

1. An electrical connection apparatus used for inspecting an object to be inspected, the electrical connection apparatus comprising:
- a terminal for inspection having a tip arranged so as to be in contact with the object to be inspected and a proximal end connected to the tip;
- a probe head holding the terminal for inspection;
- a wiring sheet with flexibility including a first connection portion arranged on a first sheet surface facing the probe head and a second connection portion arranged on a second sheet surface facing in an opposite direction of the first sheet surface; and
- a wiring board that is arranged facing the second sheet surface and includes a first electrode which is electrically connected to the second connection portion, wherein
- a press-fit pin has a head that elastically deforms in a direction perpendicular to an axial direction and a main body that is embedded in the wiring sheet, the head being exposed from at least any of the first sheet surface and the second sheet surface,
- the head of the press-fit pin exposed from the first sheet surface is fitted into a recess formed in the probe head to fix the probe head to the wiring sheet, and
- the head of the press-fit pin exposed from the second sheet surface is fitted into a recess formed in the wiring board to fix the wiring board to the wiring sheet.

2. The electrical connection apparatus according to claim **1**, wherein
- the head is provided in plurality, and the press-fit pin has the main body passing through the wiring sheet, from the first sheet surface to the second sheet surface,
- a first head of the heads connected to one end of the main body is exposed from the first sheet surface, and
- a second head of the heads connected to the other end of the main body is exposed from the second sheet surface.

3. The electrical connection apparatus according to claim **1**, wherein
- the head connected to the main body of the press-fit pin is exposed only from either the first sheet surface or the second sheet surface.

4. The electrical connection apparatus according to claim **1**, wherein
- the press-fit pin is a metal material.

5. The electrical connection apparatus according to claim **1**, wherein the head is star-shaped when viewed from the axial direction of the press-fit pin.

6. The electrical connection apparatus according to claim **1**, wherein

an outer edge of the wiring board is positioned outward from an outer edge of the wiring sheet in plan view.

7. The electrical connection apparatus according to claim **1**, wherein

a maximum diameter of the head exposed from the first sheet surface is different from a maximum diameter of the head exposed from the second sheet surface.

8. The electrical connection apparatus according to claim **1**, wherein

an inner diameter of the recess formed in the probe head is different from an inner diameter of the recess formed in the wiring board.

9. The electrical connection apparatus according to claim **1**, wherein

the wiring sheet has a structure in which a laminated body constituted by a conductive film and an insulating film is interposed between cover films of an insulating material.

10. The electrical connection apparatus according to claim **1**, wherein

the wiring board includes a second electrode electrically connected to the first electrode via internal wiring, the electrical connection apparatus further comprising:

a printed board that is arranged facing the wiring board and has a wiring pattern connected to the second electrode.

11. The electrical connection apparatus according to claim **10**, wherein

the terminal for inspection is provided in plurality, the wiring pattern is provided in plurality, and the wiring board is a space transformer that transforms a distance between the terminals for inspection into a distance between the wiring patterns of the printed board, when viewed from a direction normal to a main surface of the wiring board.

12. The electrical connection apparatus according to claim **1**, wherein

the terminal for inspection is a probe having a tip that is arranged so as to be in contact with a signal terminal of the object to be inspected and a proximal end connected to the first connection portion of the wiring sheet.

13. The electrical connection apparatus according to claim **1**, wherein

the terminal for inspection is arranged in a test socket connected to an external terminal of a package in which the object to be inspected is mounted.

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